

NYDA, Russian Federation

WMO: 233450

Lat: 66.624N

Lon: 72.896E

Elev: 7

StdP: 101.24

Time Zone: 5.00 (E05)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.8	-40.1	-45.6	0.0	-42.3	-43.1	0.1	-39.8	15.6	-13.5	13.3	-12.1	2.8	130	0.942

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.7	26.8	18.3	24.4	17.6	21.4	15.9	19.3	24.2	17.9	23.2	16.6	21.0	5.3	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.2	12.3	22.2	15.8	11.2	20.5	14.4	10.2	19.0	54.9	24.7	50.7	23.3	46.7	21.1	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.2	12.3	11.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-44.2	20.7	1.8	1.9	-45.5	22.0	-46.6	23.1	-47.6	24.1	-48.9	25.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-5.6	-24.8	-21.7	-15.4	-8.7	-2.3	10.0	15.6	11.5	5.6	-3.9	-15.4	-19.0		
	DBStd	14.99	9.05	9.21	9.29	6.26	5.15	6.06	4.51	3.29	3.69	5.21	7.65	9.15		
	HDD10.0	6033	1078	887	788	561	383	78	4	21	140	432	761	898		
	HDD18.3	8778	1337	1120	1047	811	641	256	112	213	383	691	1011	1157		
	CDD10.0	331	0	0	0	0	1	77	178	67	7	0	0	0		
	CDD18.3	33	0	0	0	0	0	5	27	1	0	0	0	0		
	CDH23.3	308	0	0	0	0	0	47	254	8	0	0	0	0		
	CDH26.7	62	0	0	0	0	0	6	56	0	0	0	0	0		
(14) Wind	WSAvg	5.5	5.3	5.3	5.9	5.8	6.1	5.5	5.2	5.3	5.5	5.4	5.5	5.3		
(15) Precipitation	PrecAvg	353	16	14	17	22	27	38	50	49	40	37	23	20		
	PrecMax	489	39	65	40	72	53	79	134	79	67	72	46	48		
	PrecMin	201	6	6	2	0	8	11	3	18	9	20	9	7		
	PrecStd	70	7	11	10	15	12	16	31	17	15	12	9	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.7	-2.4	1.7	3.7	16.4	27.4	30.2	24.6	18.9	8.1	1.7	-0.9		
		MCWB	-3.0	-3.1	0.7	1.5	11.6	18.4	19.3	16.8	13.3	7.3	1.1	-1.4		
	2%	DB	-6.8	-4.7	0.2	2.3	10.3	24.6	28.2	21.3	14.7	5.2	-0.3	-3.0		
		MCWB	-7.3	-5.4	-0.6	0.9	7.0	16.5	18.6	15.6	11.6	4.0	-0.7	-3.7		
	5%	DB	-9.0	-6.6	-0.9	1.5	6.1	21.7	26.4	18.7	12.9	3.5	-2.1	-4.7		
		MCWB	-9.4	-7.2	-1.8	0.5	4.0	14.9	18.3	14.6	10.4	2.4	-2.6	-5.1		
	10%	DB	-11.9	-9.0	-2.3	0.6	3.2	19.5	23.9	16.9	11.0	2.0	-4.6	-6.4		
		MCWB	-12.3	-9.6	-3.1	-0.2	1.8	14.7	18.0	13.5	9.3	1.2	-5.1	-6.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.8	-2.9	0.7	2.0	11.7	19.5	21.7	18.0	14.0	7.3	1.0	-1.3		
		MCDB	-2.5	-2.3	1.4	3.0	16.5	25.0	28.1	22.2	17.7	8.4	1.5	-1.1		
	2%	WB	-7.2	-5.3	-0.5	1.2	7.6	18.0	20.4	16.6	12.2	4.3	-0.7	-3.5		
		MCDB	-6.9	-4.7	0.2	2.0	10.3	22.5	25.8	20.3	14.4	4.9	-0.3	-3.0		
	5%	WB	-9.5	-7.2	-1.7	0.6	4.2	16.2	19.0	15.1	10.7	2.5	-2.6	-5.1		
		MCDB	-9.1	-6.6	-1.0	1.5	6.0	20.6	24.1	18.3	12.4	3.3	-2.1	-4.7		
	10%	WB	-12.3	-9.6	-3.0	-0.2	2.0	14.5	17.8	13.7	9.4	1.2	-5.0	-6.9		
		MCDB	-12.0	-9.1	-2.3	0.7	3.0	19.1	23.4	16.2	10.9	1.9	-4.6	-6.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	8.0	8.6	9.0	6.0	7.7	7.7	6.2	5.2	4.9	7.1	7.8		
		MCDBR	10.2	8.7	7.6	7.4	10.1	12.9	12.9	10.4	9.0	5.5	5.3	7.3		
	5% WB	MCWBR	10.1	8.6	7.3	6.5	7.0	6.6	5.7	5.7	5.9	4.2	5.3	7.3		
		MCWBR	10.1	8.9	8.0	6.4	9.5	11.8	11.6	8.8	7.9	4.8	5.7	7.1		
(40) Clear-Sky Solar Irradiance	taub		0.211	0.262	0.312	0.331	0.350	0.370	0.395	0.374	0.325	0.281	0.205	N/A		
		taud	1.871	2.011	2.015	2.056	2.151	2.344	2.324	2.362	2.447	2.214	1.923	N/A		
	Ebn at Noon		160	555	713	813	840	835	796	773	736	543	159	0		
		Edh at Noon	16	58	98	122	124	106	106	92	66	47	14	0		
(44) All-Sky Solar Radiation	RadAvg		0.06	0.57	2.19	4.32	5.71	5.33	4.92	3.32	1.82	0.66	0.13	0.00		
	RadStd		0.01	0.07	0.12	0.26	0.28	0.41	0.64	0.37	0.23	0.10	0.02	0.00		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page